Practice: two budget lines, substitutes and complements

Lecture notes

What this lecture is about. Lecture 13 practised the optimal bundle on budgets made of two straight segments. Two such budgets are used throughout, built from the same two lines, $2x + y = 14$ and $x + y = 10$, which cross at $(4, 6)$. On *budget A* the consumer may choose either line, so the budget is the outer boundary and the kink points towards the origin. On *budget B* both limits must hold, so the budget is the inner boundary and the kink points away from the origin. Questions 1 and 2 use perfect substitutes; questions 3 and 4 use perfect complements. Question 1 ends with one Cobb-Douglas case.

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1 The two budgets and the method

Budget A: either line can be chosen

$$2x + y = 14 \ \text{ for } 0 \le x \le 4, \qquad x + y = 10 \ \text{ for } 4 \le x \le 10.$$

The consumer may use either of the two lines, so every bundle below at least one of them is affordable. The budget line is the outer boundary: the steeper line on the left, the flatter line on the right. The kink $(4, 6)$ points towards the origin.

Budget B: both limits hold

$$x + y = 10 \ \text{ for } 0 \le x \le 4, \qquad 2x + y = 14 \ \text{ for } 4 \le x \le 7.$$

Both limits must hold at once, so only bundles below both lines are affordable. The budget line is the inner boundary: the flatter line on the left, the steeper line on the right. The kink $(4, 6)$ points away from the origin.

On each segment, p_x is the number of units of y given up for one more unit of x: $p_x = 2$ on $2x + y = 14$ and $p_x = 1$ on $x + y = 10$; in both cases $p_y = 1$.

The comparison. At a bundle on a segment, compare MU_x/p_x (the extra utility from the last euro spent on x, called the *bang for the buck* in class) with MU_y/p_y, using that segment's prices. A higher ratio for x means: move towards more x; a higher ratio for y: towards less x; equal ratios: stop. At the kink, compare the ratios with the prices just before and just after it.

The pictures. x is on the horizontal axis and y on the vertical axis. The budget line ("composite budget line") is orange; thin orange lines continue each segment beyond the kink. The blue line or curve is the indifference curve through the optimal bundle. Arrows show the direction of the comparison; a label such as "1: at $(4, 6)$: $3/2$ vs 1" gives the two ratios at that bundle on segment 1. A dashed circle marks a point that lies off its segment. In questions 3 and 4 a dashed blue ray from the origin, labelled "kinks", joins the corners of the L-shaped indifference curves.

2 Question 1: perfect substitutes on budget A

With $u = ax + by$ the marginal utilities are constant, $\text{MU}_x = a$ and $\text{MU}_y = b$. On each segment the comparison therefore gives the same answer at every point, and the consumer moves to an end of the segment.

2.1 Part 1: $u = 3x + y$

Question 1, part 1

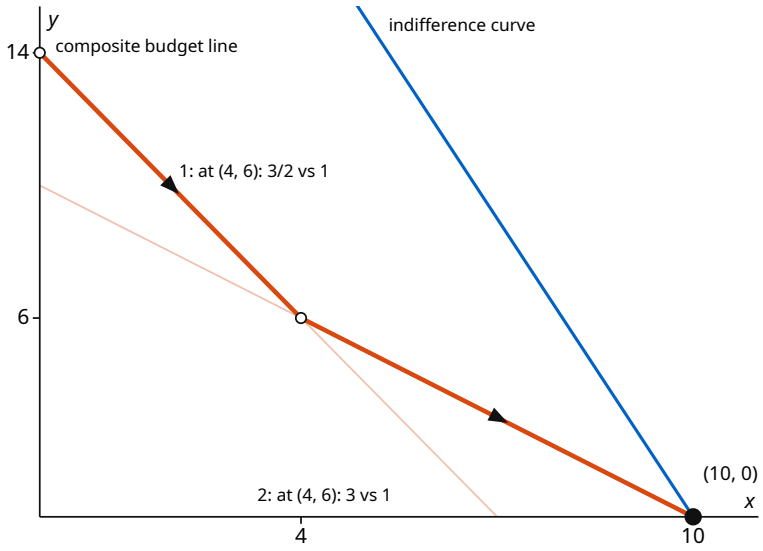
The utility function is $u = 3x + y$ and the budget is budget A. Find the optimal bundle.

Segment 1, $2x + y = 14$**.** The ratios are $3/2$ for x against 1 for y at every point: move right, to the kink $(4, 6)$.

Segment 2, $x + y = 10$**.** The ratios are 3 against 1 at every point: move right, to the end $(10, 0)$.

The kink $(4, 6)$**.** Before it, $3/2$ against 1; after it, 3 against 1. Both say more x, so the consumer passes the kink and continues to $(10, 0)$, where y cannot fall further.

The optimal bundle is the corner $(10, 0)$, with $u = 30$.



Question 1, part 1: the optimal bundle $(10, 0)$.

2.2 Part 2: $u = x + 4y$

Question 1, part 2

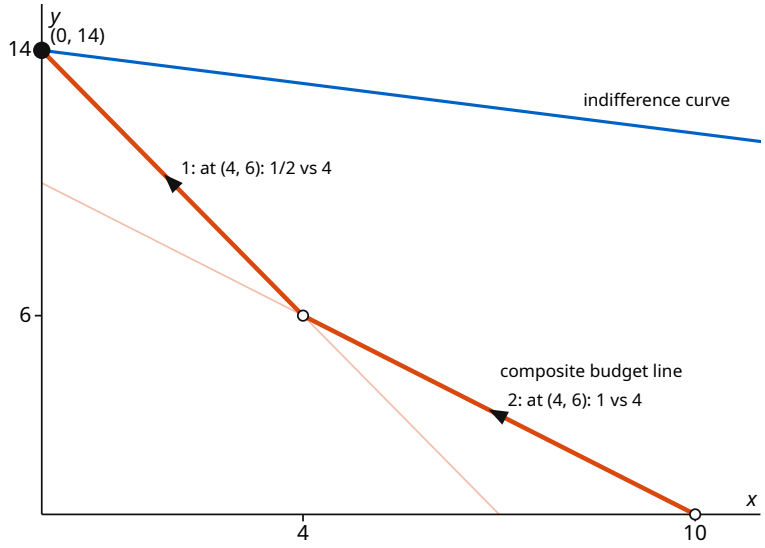
The budget is still budget A; the utility function is now $u = x + 4y$. Find the optimal bundle.

Segment 1. $1/2$ for x against 4 for y at every point: move left, to $(0, 14)$.

Segment 2. 1 against 4 at every point: move left, to the kink $(4, 6)$.

The kink. Before it, $1/2$ against 4; after it, 1 against 4. Both say less x, so the consumer passes the kink and continues to $(0, 14)$.

The optimal bundle is the corner $(0, 14)$, with $u = 4 \cdot 14 = 56$.



Question 1, part 2: the optimal bundle $(0, 14)$.

2.3 Part 3: $u = 7x + 5y$

Question 1, part 3

The budget is still budget A; the utility function is now $u = 7x + 5y$. Find the optimal bundle.

Segment 1. $7/2$ for x against 5 for y at every point: move left, to $(0, 14)$.

Segment 2. 7 against 5 at every point: move right, to $(10, 0)$.

The kink. Before it the comparison says less x; after it, more x. It pushes away from the kink in both directions: the kink is worse than the bundles on either side of it.

Compare the two ends. The comparison stops at two bundles, so their utilities must be compared:

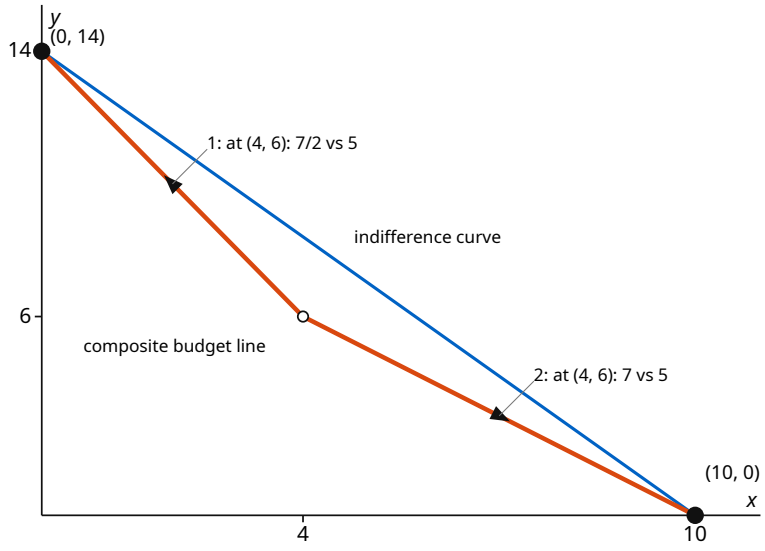
$$u(0, 14) = 5 \cdot 14 = 70, \qquad u(10, 0) = 7 \cdot 10 = 70.$$

The kink gives only $u(4, 6) = 28 + 30 = 58$.

Both $(0, 14)$ and $(10, 0)$ are optimal, each with $u = 70$.

Insight: the kink of budget A

On budget A the kink lies below the straight line joining the two ends. With perfect substitutes, utility along that straight line changes at a constant rate, so one of the two ends is always at least as good as the kink. The kink is never the only optimal bundle here; the third case on budget A is a tie between the two ends.



Question 1, part 3: the two optimal bundles $(0, 14)$ and $(10, 0)$.

2.4 Part 4: $u = xy$

Question 1, part 4

The budget is still budget A; the utility function is now $u = xy$. Find the optimal bundle.

Marginal utilities. $\mathrm{MU}_x = y$ and $\mathrm{MU}_y = x$. On a segment where one unit of x costs p_x (and $p_y = 1$), equal ratios give $y/p_x = x$, that is $y = p_x x$: the consumer spends half of the segment's budget on each good.

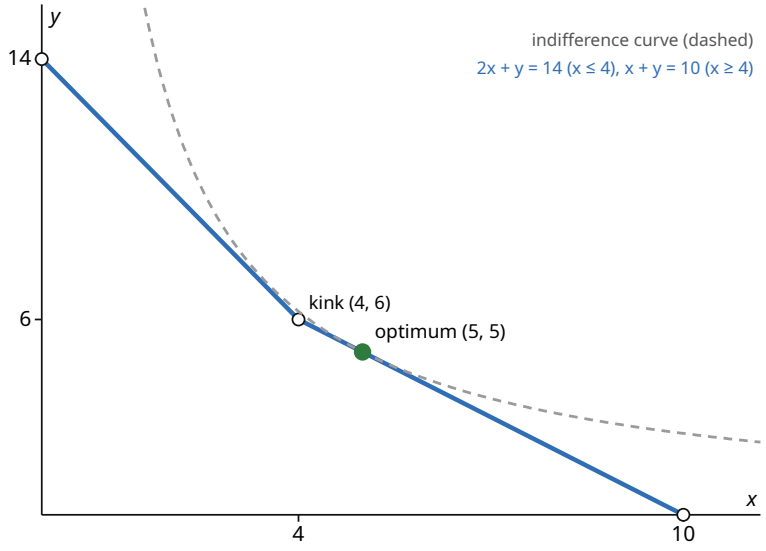
Segment 1, $2x + y = 14$**.** $y = 2x$ gives $4x = 14$, so $x = \frac{7}{2}$ and $y = 7$. Since $\frac{7}{2} \leq 4$, this point lies on the segment: $u = \frac{7}{2} \cdot 7 = \frac{49}{2} = 24.5$.

Segment 2, $x + y = 10$**.** $y = x$ gives $2x = 10$, so $x = 5$ and $y = 5$. Since $5 \geq 4$, this point lies on the segment: $u = 25$.

The kink. $u(4, 6) = 24$.

Compare. Each segment has its own stopping point, so the utilities are compared: $25 > 24.5 > 24$.

The optimal bundle is $(5, 5)$, with $u = 25$. It beats $(\frac{7}{2}, 7)$ with 24.5 and the kink with 24.



Question 1, part 4: the optimal bundle $(5, 5)$. In this picture the budget line is blue and the indifference curve $xy = 25$ is dashed grey.

3 Question 2: perfect substitutes on budget B

3.1 Part 1: $u = 4x + y$

Question 2, part 1

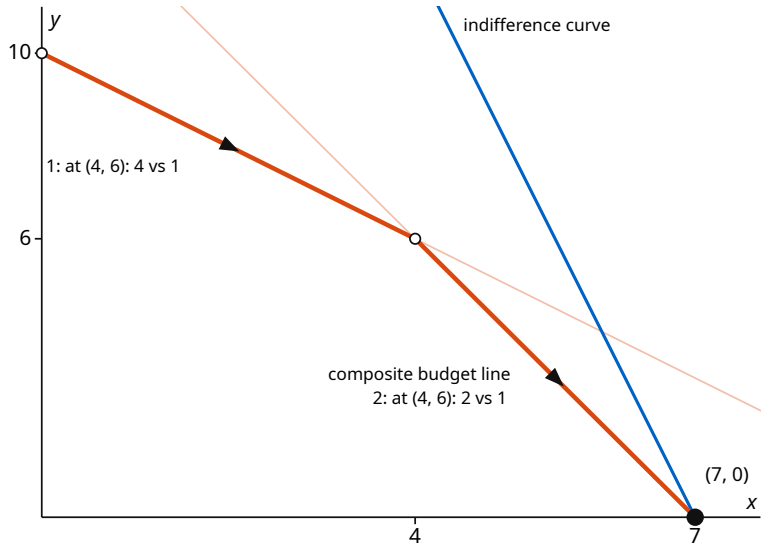
The utility function is $u = 4x + y$ and the budget is budget B. Find the optimal bundle.

Segment 1, $x + y = 10$**.** 4 for x against 1 for y at every point: move right, to the kink $(4, 6)$.

Segment 2, $2x + y = 14$**.** $4/2 = 2$ against 1 at every point: move right, to $(7, 0)$.

The kink. Both sides say more x, so the consumer passes the kink and continues to $(7, 0)$.

The optimal bundle is the corner $(7, 0)$, with $u = 28$.



Question 2, part 1: the optimal bundle $(7, 0)$.

3.2 Part 2: $u = x + 3y$

Question 2, part 2

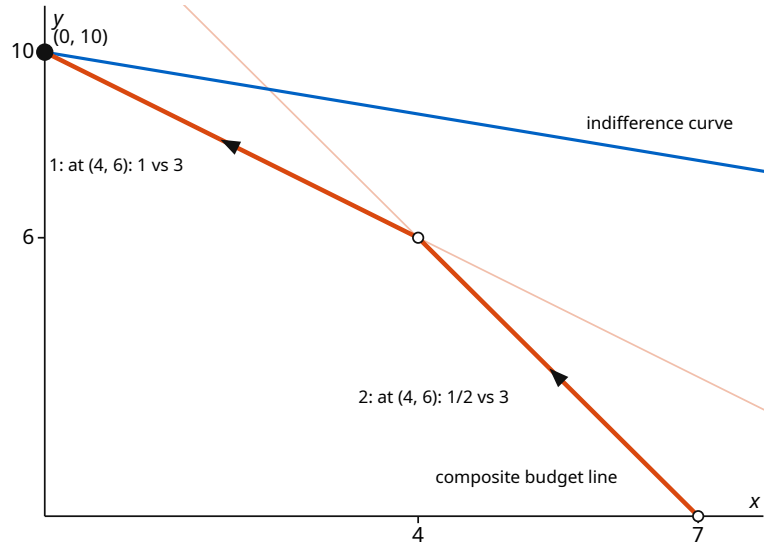
The budget is still budget B; the utility function is now $u = x + 3y$. Find the optimal bundle.

Segment 1. 1 for x against 3 for y at every point: move left, to $(0, 10)$.

Segment 2. $1/2$ against 3 at every point: move left, to the kink $(4, 6)$.

The kink. Both sides say less x, so the consumer passes the kink and continues to $(0, 10)$.

The optimal bundle is the corner $(0, 10)$, with $u = 30$.



Question 2, part 2: the optimal bundle $(0, 10)$.

3.3 Part 3: $u = 3x + 2y$

Question 2, part 3

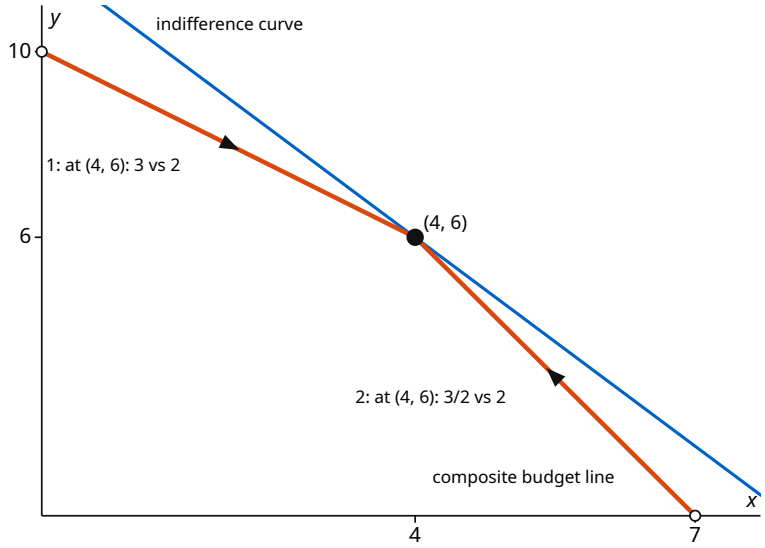
The budget is still budget B; the utility function is now $u = 3x + 2y$. Find the optimal bundle.

Segment 1, $x + y = 10$**.** 3 for x against 2 for y at every point: move right, to the kink $(4, 6)$.

Segment 2, $2x + y = 14$**.** 3/2 against 2 at every point: move left, to the kink $(4, 6)$.

The kink. Before it, more x; after it, less x. The consumer stops at the kink.

The optimal bundle is the kink $(4, 6)$, with $u = 12 + 12 = 24$.



Question 2, part 3: the optimal bundle is the kink $(4, 6)$.

4 Question 3: perfect complements on budget A

Perfect complements

With $u = \min\{ax, by\}$ the goods are used in the fixed ratio $ax = by$. The indifference curves are L-shaped, with their corners on the line $ax = by$. Away from that line, one good is in excess and adding more of it does not raise utility: its marginal utility is 0. At the corner of the L the comparison is not defined, because the marginal utilities jump there.

The method. The corners of all the L-shaped indifference curves lie on one ray from the origin, $ax = by$, that is $y = \frac{a}{b}x$. In the pictures this ray is drawn dashed and labelled "kinks". Away from the ray the comparison always says: move towards it. So the optimal bundle is where the ray of kinks meets the budget line. On each segment, find where the ray meets the segment's line and check whether that point lies on the segment. If it does not, the ray passes beyond the segment: move along the segment towards the ray, to the segment's end, here the kink $(4, 6)$.

4.1 Part 1: $u = \min\{x, y\}$

Question 3, part 1

The utility function is $u = \min\{x, y\}$ and the budget is budget A. Find the optimal bundle.

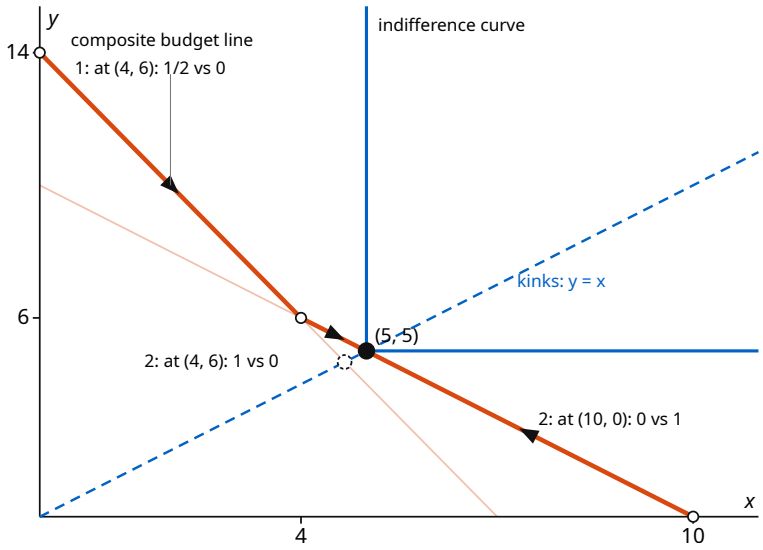
The ray of kinks. The corners of the L-shaped indifference curves lie on the ray $y = x$ (dashed in the picture). The optimal bundle is where this ray meets the budget line: at $(5, 5)$, as the steps below confirm.

Segment 1, $2x + y = 14$**.** The ray of kinks is $y = x$. Substituting gives $3x = 14$, so $x = \frac{14}{3} > 4$: off this segment. Along the segment, for example at $(0, 14)$ and at $(4, 6)$, x is the scarce good: $\text{MU}_x = 1$ and $\text{MU}_y = 0$, so the ratios are $1/2$ against 0. Move right, to the kink $(4, 6)$.

Segment 2, $x + y = 10$**.** $y = x$ gives $2x = 10$, so $x = y = 5$: on this segment. At $(4, 6)$ the ratios are 1 against 0 (more x); at $(10, 0)$ they are 0 against 1 (less x).

Where it stops. The ray of kinks meets the budget at one point only, $(5, 5)$.

The optimal bundle is $(5, 5)$, with $u = 5$.



Question 3, part 1: the optimal bundle $(5, 5)$.

4.2 Part 2: $u = \min\{5x, y\}$

Question 3, part 2

The budget is still budget A; the utility function is now $u = \min\{5x, y\}$. Find the optimal bundle.

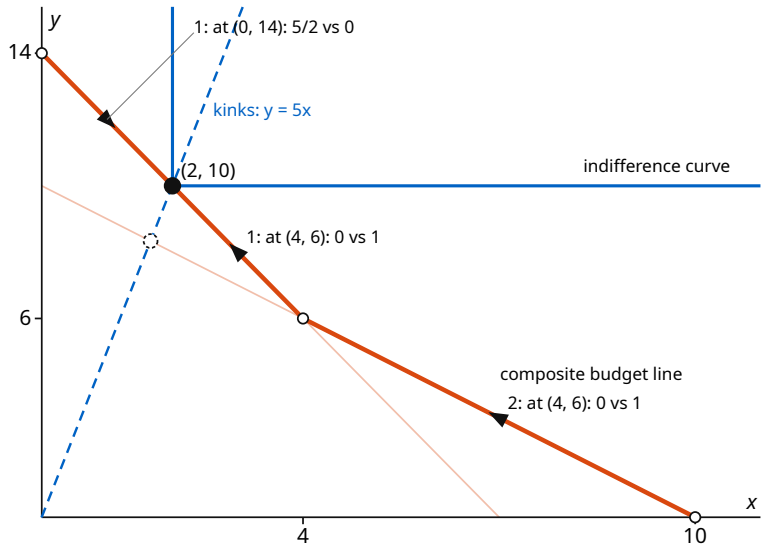
The ray of kinks. The corners of the L-shaped indifference curves lie on the ray $y = 5x$ (dashed in the picture). The optimal bundle is where this ray meets the budget line: at $(2, 10)$, as the steps below confirm.

Segment 1, $2x + y = 14$. The ray of kinks is $y = 5x$. Substituting gives $7x = 14$, so $x = 2$ and $y = 10$: on this segment. At $(0, 14)$ the ratios are $5/2$ against 0 (more x); at $(4, 6)$ they are 0 against 1 (less x).

Segment 2, $x + y = 10$. $y = 5x$ gives $6x = 10$, so $x = \frac{5}{3} < 4$: off this segment. At $(4, 6)$, y is the scarce good: 0 against 1, so move left, to the kink.

Where it stops. The ray of kinks meets the budget at one point only, $(2, 10)$.

The optimal bundle is $(2, 10)$, with $u = \min\{10, 10\} = 10$.



Question 3, part 2: the optimal bundle $(2, 10)$.

4.3 Part 3: $u = \min\{3x, 2y\}$

Question 3, part 3

The budget is still budget A; the utility function is now $u = \min\{3x, 2y\}$. Find the optimal bundle.

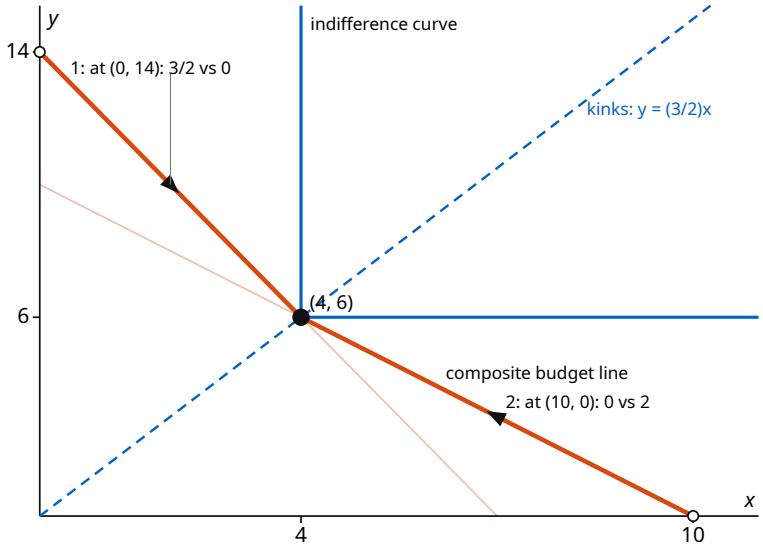
The ray of kinks. The corners of the L-shaped indifference curves lie on the ray $y = \frac{3}{2}x$ (dashed in the picture). The optimal bundle is where this ray meets the budget line: at $(4, 6)$, exactly the kink, as the steps below confirm.

Segment 1, $2x + y = 14$**.** The ray of kinks is $y = \frac{3}{2}x$. Substituting gives $\frac{7}{2}x = 14$, so $x = 4$ and $y = 6$: the kink, which belongs to this segment. At $(0, 14)$ the ratios are $3/2$ against 0 (more x).

Segment 2, $x + y = 10$**.** $y = \frac{3}{2}x$ gives $\frac{5}{2}x = 10$, so again $x = 4$ and $y = 6$. At $(10, 0)$ the ratios are 0 against 2 (less x).

Where it stops. The ray of kinks passes exactly through the kink.

The optimal bundle is the kink $(4, 6)$, with $u = \min\{12, 12\} = 12$.



Question 3, part 3: the optimal bundle is the kink $(4, 6)$.

5 Question 4: perfect complements on budget B

5.1 Part 1: $u = \min\{x, 3y\}$

Question 4, part 1

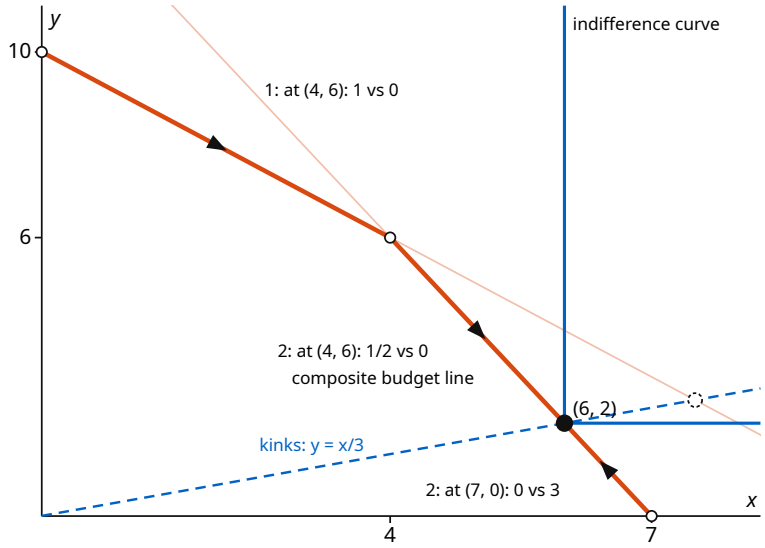
The utility function is $u = \min\{x, 3y\}$ and the budget is budget B. Find the optimal bundle.

The ray of kinks. The corners of the L-shaped indifference curves lie on the ray $y = \frac{1}{3}x$ (dashed in the picture). The optimal bundle is where this ray meets the budget line: at $(6, 2)$, as the steps below confirm.

Segment 1, $x + y = 10$**.** The ray of kinks is $y = \frac{1}{3}x$. Substituting gives $\frac{4}{3}x = 10$, so $x = \frac{15}{2} > 4$: off this segment. At $(0, 10)$ the ratios are 1 against 0: move right, to the kink $(4, 6)$.

Segment 2, $2x + y = 14$**.** $y = \frac{1}{3}x$ gives $\frac{7}{3}x = 14$, so $x = 6$ and $y = 2$: on this segment. At $(4, 6)$ the ratios are $1/2$ against 0 (more x); at $(7, 0)$ they are 0 against 3 (less x).

The optimal bundle is $(6, 2)$, with $u = \min\{6, 6\} = 6$.



Question 4, part 1: the optimal bundle $(6, 2)$.

5.2 Part 2: $u = \min\{7x, 3y\}$

Question 4, part 2

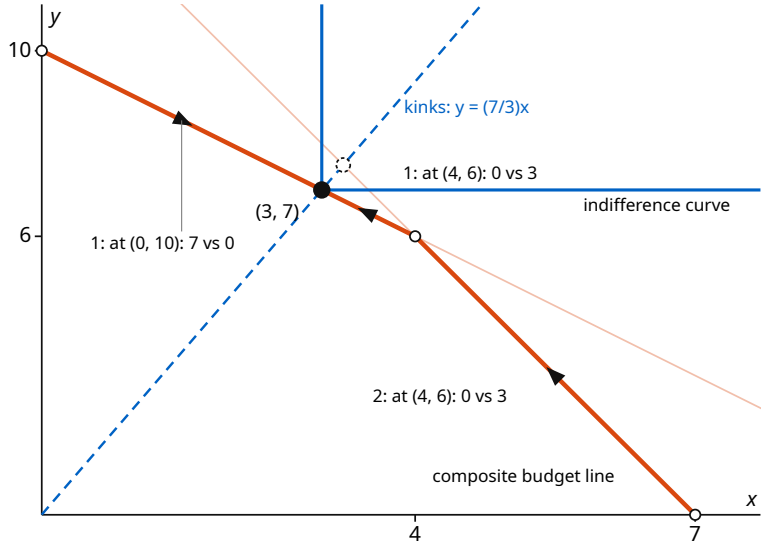
The budget is still budget B; the utility function is now $u = \min\{7x, 3y\}$. Find the optimal bundle.

The ray of kinks. The corners of the L-shaped indifference curves lie on the ray $y = \frac{7}{3}x$ (dashed in the picture). The optimal bundle is where this ray meets the budget line: at $(3, 7)$, as the steps below confirm.

Segment 1, $x + y = 10$**.** The ray of kinks is $y = \frac{7}{3}x$. Substituting gives $\frac{10}{3}x = 10$, so $x = 3$ and $y = 7$: on this segment. At $(0, 10)$ the ratios are 7 against 0 (more x); at $(4, 6)$ they are 0 against 3 (less x).

Segment 2, $2x + y = 14$**.** $y = \frac{7}{3}x$ gives $\frac{13}{3}x = 14$, so $x = \frac{42}{13} < 4$: off this segment. At $(4, 6)$ the ratios are 0 against 3: move left, to the kink.

The optimal bundle is $(3, 7)$, with $u = \min\{21, 21\} = 21$.



Question 4, part 2: the optimal bundle $(3, 7)$.

5.3 Part 3: $u = \min\{3x, 2y\}$

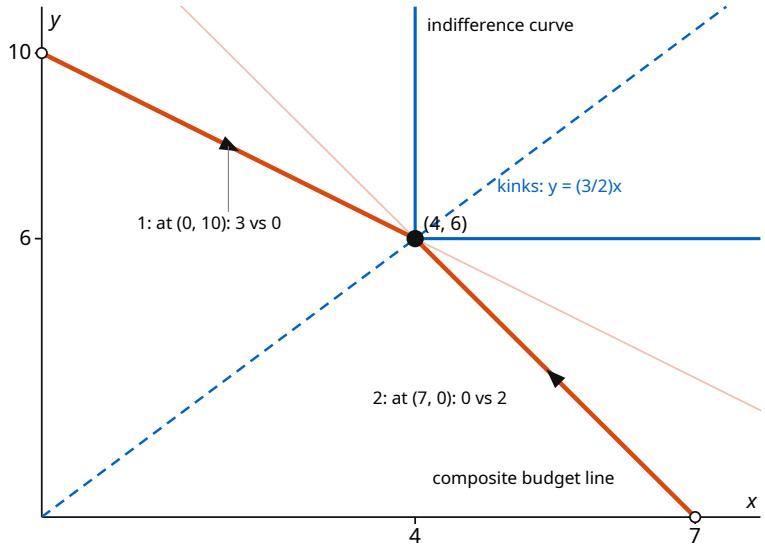
Question 4, part 3

The budget is still budget B; the utility function is now $u = \min\{3x, 2y\}$. Find the optimal bundle.

The ray of kinks. The corners of the L-shaped indifference curves lie on the ray $y = \frac{3}{2}x$ (dashed in the picture). The optimal bundle is where this ray meets the budget line: at $(4, 6)$, exactly the kink, as the steps below confirm.

Both segments. The ray of kinks $y = \frac{3}{2}x$ meets $x + y = 10$ at $\frac{5}{2}x = 10$ and $2x + y = 14$ at $\frac{7}{2}x = 14$: both give $(4, 6)$, the kink. At $(0, 10)$ the ratios are 3 against 0 (more x); at $(7, 0)$ they are 0 against 2 (less x).

The optimal bundle is the kink $(4, 6)$, with $u = 12$.



Question 4, part 3: the optimal bundle is the kink $(4, 6)$.

6 Summary

question, part	budget	utility	optimal bundle
1.1	A	$3x + y$	$(10, 0)$
1.2	A	$x + 4y$	$(0, 14)$
1.3	A	$7x + 5y$	$(0, 14)$ and $(10, 0)$
1.4	A	xy	$(5, 5)$
2.1	B	$4x + y$	$(7, 0)$
2.2	B	$x + 3y$	$(0, 10)$
2.3	B	$3x + 2y$	kink $(4, 6)$
3.1	A	$\min\{x, y\}$	$(5, 5)$
3.2	A	$\min\{5x, y\}$	$(2, 10)$
3.3	A	$\min\{3x, 2y\}$	kink $(4, 6)$
4.1	B	$\min\{x, 3y\}$	$(6, 2)$
4.2	B	$\min\{7x, 3y\}$	$(3, 7)$
4.3	B	$\min\{3x, 2y\}$	kink $(4, 6)$

- Perfect substitutes: on each segment the comparison is the same everywhere, so the answer is an end of a segment. On budget B the kink can be optimal; on budget A it cannot be the only optimum, and the third case is a tie between the two ends.
- Perfect complements: the optimal bundle lies where the ray of kinks $ax = by$ meets the budget line, on one segment or exactly at the kink, on either budget.
- With a kink pointing towards the origin (budget A), each segment can have its own stopping point; compare their utilities.